MATH 128a: Homework #4

Instructor: F. Weber

Due by: **Wednesday, October 1, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from textbook Section 2.1

Solve problems 19(17) from Section 2.1 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.

Problem 2: Problems from textbook Section 2.2

Solve problems 1d, 10(8), 13ab(11ab), 21(17), 22(18) from Section 2.2 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.
13a(11a): use the starting interval [a,b]=[0,1].
13b(11b): use the starting interval [a,b]=[2.5,3].
22(18): the 9th edition book has two typos:
Part (a) should refer to Theorem 2.3.
Part (b) should refer to Theorem 2.4.

Problem 3: Consistent fixed point formulations

The nonlinear equation

$$f(x) := xe^x - 1 = 0 \tag{1}$$

has the solution

$$x^* = 0.56714329040978\ldots$$

Consistent fixed point formulations of $f(x) = 0$ satisfy $G(x) = x$.
Consider the following consistent fixed point formulations of (1):

$$G_1(x) = e^{-x},$$
$$G_2(x) = \frac{1+x}{1+e^x},$$
$$G_3(x) = x + 1 - xe^x.$$

Compute 10 iterations for each of these formulations using the nonlinear fixed point iteration starting from value $x_0 = 0.5$. Measure the error to the fixed point x^*. What do you observe? Which reformulations converge and which do not converge? Explain why some of them converge faster/slower or why they don't converge.
You may use the Matlab scripts and functions available on the bcourses page.